

UMESH SHARMA

MS, Transportation Engineering

umesh.sharma@okstate.edu · +1 (405) 269-1756 · Stillwater, OK

SUMMARY

Graduate Transportation Engineer with 2+ years of experience in transportation and traffic engineering, roadway design, and state-wide network optimization. My strengths lie in traffic engineering, ITS, GIS-based analysis, and data-driven planning.

EDUCATION

MS Civil Engineering (Transportation), *Oklahoma State University* · GPA: 4.0/4.0 Aug 2024 – Aug 2026

Thesis: Evaluating and Optimizing Statewide Traffic Monitoring and HPMS Program in Oklahoma Stillwater, OK

Coursework: Transportation Safety & Analysis, Advanced Transportation Systems, Civil Infrastructure Systems, Project Management

BE Civil Engineering, *Tribhuvan University* Nov 2018 – Apr 2022

Coursework: Urban & Regional Transportation Planning, Project Engineering Kathmandu, Nepal

SKILLS & COMPETENCIES

Technical: Traffic & ITS Engineering | HPMS Evaluation | Traffic Signal Timing/Design | Safety (HSM/HSIP) | PS&E

Software: AutoCAD | Civil 3D | MicroStation | OpenRoads | ArcGIS Pro | PTV VISSIM | SUMO | MS Project | MS Office Suite

Data & Analysis: Python | GIS Spatial Analysis | Machine Learning | Visualization | Excel (advanced)

Standards: HPMS Field Manual | FHWA TMG | HCM | HSM | MUTCD | ADA/PROWAG

EXPERIENCE

Graduate Research Assistant, *Oklahoma State University* Aug 2024 – Present
Stillwater, OK

- Optimized statewide 18,783 Short-Term Count sites by removing redundancy using spatial, geographic, and roadway barriers, cutting traffic count program costs by 21% (\$140K/yr) and in compliance with FHWA Traffic Monitoring Guide (TMG).
- Redesigned AADT adjustment factors via K-means clustering across 200+ Continuous Count Stations using historical data.
- Improved statewide data efficiency by integrating ODOT and MPO datasets, eliminating redundant counts, and strengthening regional coordination for 6,000+ roadway sites
- Designed CV-based STC stability classification for a traffic count cycle projected to reduce 27 % reduction in field data collection workload while maintaining HPMS data quality thresholds.
- Supported dynamic impact factor evaluation for I-35 bridges through integrated sensor-based field data collection, applying machine learning to analyze vehicle-pavement interaction patterns.

Associate Civil Engineer, *Engineering Helpline Pvt. Ltd.* Nov 2022 – May 2024
Kathmandu, Nepal

- Contributed to detailed engineering design and cost estimate report for 14-mile, \$8.5M roadway project, developing 50+ engineering drawings, including alignment, drainage, cross-sections, and comprehensive budget analysis that secured project approval.
- Supervised 10+ construction sites; enforced geometric design standards and QC protocols (15% defect reduction); produced roadway plan sets using AutoCAD and Civil 3D with multidisciplinary coordination.

Intern Civil Engineer, *SunFarmer Nepal* Jun 2022 – Oct 2022
Kathmandu, Nepal

- Conducted site assessments, feasibility studies, and GIS-based spatial analysis for solar infrastructure projects; prepared technical reports and coordinated cross-functional stakeholder communication.

CERTIFICATIONS & TRAINING

Fundamentals of Engineering (FE Civil) | Licensed Engineer, Nepal Engineering Council | Python for Data Science & ML | Going Places with Spatial Analysis (ESRI) | Writing Skills for Engineering Leaders

LEADERSHIP

President, ITE Student Chapter, OSU: Revamped dormant ITE chapter and led statewide transportation symposium, attracting 150+ participants across 10+ technical sessions.

Co-Founder, The Mind Workshop: Managed merchandise logistics & marketing for a hybrid inventory model bookstore.